Convex Optimization and Approximation Algorithms

Problem Set 4 – CS 6515/4540 (Fall 2026)

This problem set is due on **Sunday, October 25th**. Submission is via Gradescope. Your solution must be a typed PDF (e.g. via LaTeX); no handwritten solutions.

16. Non-Convex Function

In class we assumed function f is convex. This problem considers the non-convex case.

We want to show that for β-smooth f, after T iterations with step size $\eta \leq \frac{1}{\beta}$ we can find a point x' with

$$\|\nabla f(x')\| \leq \sqrt{\left(\frac{2}{\eta T}\right)(f(x^0) - f(x^*))}.$$

(Note that for a local optimum we have $\nabla f(x) = 0$, so a small norm $\|\nabla f(x')\|$ indicates that we are close to a local optimum or saddle point.)

Proving this from scratch is a bit tricky, so we provide the following subproblems to guide you to a proof. Each subproblem can be solved in a few lines of calculation/algebra.

Problem:

1. Show

$$f(x^{t+1}) \leq f(x^t) - \left(\frac{\eta}{2}\right) \|\nabla f(x^t)\|^2.$$

   Hint: Check the proof from class. Does it work for non-convex functions?

2. Show

$$\sum_{k=0}^{T} \|\nabla f(x^k)\|^2 \leq \left(\frac{2}{\eta}\right)(f(x^0) - f(x^*)).$$

   Hint: Part 1 implies

$$\left(\frac{\eta}{2}\right)\|\nabla f(x^t)\|^2 \leq \ldots$$

3. Show

$$\min_{0 \leq k \leq T} \|\nabla f(x^k)\| \leq \sqrt{\left(\frac{2}{\eta T}\right)(f(x^0) - f(x^*))}.$$

where x^* is the global optimum satisfying

$$f(x^*) = \min_x f(x).$$

You are allowed to use subproblems to solve later subproblems (e.g., use 1 and 2 to solve 3), even if you did not prove them.

17. Greedy Approximation Algorithm

Consider placing n items into m bins. Item i has weight w_i.

We want to place these n items into the bins such that the maximum total weight of a bin is minimized.

Consider the following greedy algorithm:

Sort the weights w_i in decreasing order. Going through the items in decreasing order of weight, place each item in the bin with the least current load.

Show this algorithm gives a 1.5-approximation for the least possible maximum load of any placement of these n items into the m bins.

18. Hitting Set via Deterministic Rounding

Let

$$W = \{w_1, w_2, ..., w_n\}$$

be a set of positive numbers and let

$$B_1, B_2, ..., B_m$$

be subsets of

$$[n] = \{1, 2, ..., n\}.$$

We say that

$$H \subseteq [n]$$

is a *hitting set* if

$$H \cap B_i \neq \varnothing$$

for every $1 \leq i \leq m$.

Denote by

$$b = \max_j |B_j|$$

the size of the largest set B_j.

Consider the **Minimum Hitting Set** problem, where we assign weight w_i to element i and seek a hitting set of minimum weight

$$\sum_{i \in H} w_i.$$

1. Write an LP instance for this problem. Clearly define the variables, objective function, and constraints.
2. Let x^* be a solution of the relaxed LP problem. Consider the deterministic rounding

   $$\hat{x}_i = 1 \quad \text{if } x_i^* \geq \frac{1}{b},$$

   and

   $$\hat{x}_i = 0 \quad \text{otherwise.}$$

   Prove that

   $$H' = \{i : \hat{x}_i = 1\}$$

   is a hitting set.
3. Show that the deterministic rounding in part (b) yields a b-approximation for the **Minimum Hitting Set** problem.